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Infighting in the Dark

Multi-Label Backdoor Attack in Federated Learning

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Project Links:
Official Mirage Repository : https://github.com/NUAA-SmartSensing/Mirage
Our Project Repository : https://github.com/tanikaaaa/Mirage
Project Presentation :
https://drive.google.com/file/d/1IdUHSbUVGsi6ccXtwK88ui-Cko0k6Uyo/view?usp=drive_link

Abstract

Federated Learning (FL) is a distributed machine learning paradigm that enables multiple clients to collaboratively train a shared global model without exposing raw local data. Although FL improves privacy, its decentralized training mechanism also makes it vulnerable to backdoor attacks, where malicious participants poison local updates to implant hidden malicious behavior into the global model. Most existing research focuses on the Single-Label Backdoor Attack (SBA) setting, where all attackers target the same class. In practical federated systems, however, multiple independent adversaries may target different classes without any coordination, forming the more realistic Multi-Label Backdoor Attack (MBA) scenario.

This project implements and experimentally reproduces Mirage (Li et al., CVPR 2025), the first non-cooperative MBA framework for FL. Mirage addresses the inter-attacker exclusion problem by constructing in-distribution (ID) backdoor mappings, ensuring that poisoned samples follow the same feature-space activation pathway as clean target-class samples. Unlike traditional out-of-distribution backdoor mappings, this approach allows multiple attackers to coexist without suppressing each other's attacks. Mirage achieves this through adversarial adaptation for ID mapping construction and constrained optimization for maintaining persistence during federated training.

We reproduce the paper's experiments on CIFAR-10, CIFAR-100, and GTSRB using ResNet-18 under non-IID federated settings. On CIFAR-10, using the provided pretrained checkpoint, we achieve 92.25% clean accuracy and an average ASR of 99.84%, closely matching the reported results. Due to hardware limitations, CIFAR-100 and GTSRB were trained with abbreviated pretraining but still demonstrated successful attack behavior.

In addition, we propose Adaptive Frequency-Domain Mirage (AFDM), a novel extension that shifts trigger optimization from pixel space to the frequency domain using Discrete Cosine Transform (DCT) to improve stealthiness and resistance against spatial anomaly detection methods.

1. Introduction

Federated Learning (FL) has emerged as a promising approach to privacy-preserving collaborative machine learning. Instead of centralizing raw data, FL allows participants to train models locally and share only model updates with a central server. While this preserves data privacy, it also introduces a major security vulnerability: malicious participants can poison local updates to implant hidden behaviors into the aggregated global model.

Most existing research on federated backdoor attacks assumes the Single-Label Backdoor Attack (SBA) setting, where all attackers collaborate and target the same class. In practical large-scale FL systems, however, multiple independent adversaries may exist with different objectives and target classes, without any coordination. This forms the Multi-Label Backdoor Attack (MBA) scenario.

Simply running multiple SBA attackers simultaneously fails because different attacks compete for overlapping neural activation pathways. As established in the Mirage paper and observed during implementation, only the dominant attacker's backdoor survives aggregation, while weaker attacks are suppressed. The paper refers to this phenomenon as "infighting."

Mirage resolves this issue by replacing traditional Out-of-Distribution (OOD) mappings with In-Distribution (ID) mappings. Instead of forcing poisoned samples into specialized activation pathways, Mirage aligns poisoned samples with the clean feature-space pathway of the target class. Since clean activation pathways across classes do not compete with one another, multiple independent attackers can coexist without interfering with each other.

This project implements the full Mirage framework using the official codebase, reproduces the paper's core experimental results, and contributes an original conceptual extension. Our specific contributions are:

- Full implementation of Mirage on CIFAR-10, CIFAR-100, and GTSRB using ResNet-18 under federated non-IID settings, closely following the paper's experimental protocol.
- Experimental reproduction of the paper's core results, with CIFAR-10 results validated against the paper's figures using a provided pretrained checkpoint.
- Systematic comparison against five baseline attack methods (Vanilla, PGD, Neurotoxin, Chameleon, A3FL) in the MBA setting, with analysis of why each baseline fails.
- Evaluation against six SOTA defenses and analysis of why existing OOD-based defenses are insufficient for the MBA threat model.
- AFDM: Adaptive Frequency-Domain Mirage, a proposed extension that optimizes triggers in DCT frequency space to improve stealthiness.

2. Background and Related Work

2.1 Federated Learning and the Backdoor Threat

Federated Learning (FL) follows an iterative protocol where a central server distributes a global model to selected clients. Each client performs local training on private data and uploads model updates, which are aggregated using methods such as FedAvg [1].

Although FL preserves data privacy, it is vulnerable to backdoor attacks. Since the server receives only model updates instead of raw data, malicious clients can poison local training by inserting trigger patterns into a subset of samples. After aggregation, the global model learns to misclassify triggered inputs into attacker-defined target classes [2].

2.2 Single-Label Backdoor Attacks

Most prior work focuses on the Single-Label Backdoor Attack (SBA) setting, where all attackers share the same target class. Static-trigger methods such as Vanilla [2] and Neurotoxin [3] use fixed trigger patterns, while adaptive methods like A3FL [4], 3DFed [5], and Chameleon [6] optimize triggers dynamically for better stealth and persistence.

BackdoorIndicator [7] is a strong defense against SBA attacks because it detects the Out-of-Distribution (OOD) nature of poisoned samples. This limitation motivates the In-Distribution (ID) mapping strategy introduced by Mirage.

2.3 Backdoor Defenses in FL

Defense methods are broadly divided into detection-based and mitigation-based approaches. FoolsGold [10] reduces the weight of highly similar client updates to detect colluding attackers, while BackdoorIndicator [7] identifies poisoned models using OOD feature detection. Multi-Krum [11] and FLAME attempt to suppress poisoned updates during aggregation.

However, these defenses are primarily designed for SBA settings. FoolsGold assumes attacker collaboration, and BackdoorIndicator fails when poisoned samples follow clean target-class feature distributions, as achieved by Mirage through ID mapping.

2.4 The Multi-Label Backdoor Problem

DBA [9] distributes trigger patterns across multiple clients but still assumes a shared target class. NBA [8] considers independent attackers with different targets but does not identify the root cause of attack failure.

Mirage is the first framework to formally identify inter-attacker exclusion as competition between OOD activation pathways. It resolves this issue by constructing In-Distribution (ID) mappings, allowing multiple independent attackers to coexist without interfering with each other.

3. Methodology: The Mirage Framework

3.1 Threat Model

Each adversary is a standard FL participant. They control their own local dataset and training Each adversary behaves as a normal FL participant with control over its local dataset and training process. Attackers can access the global model broadcast each round but have no knowledge of other adversaries, their target classes, or even their existence. The server may employ any existing defense mechanism.

The objective is to implant a backdoor that causes triggered inputs to be classified as a chosen target class while maintaining high clean accuracy to avoid detection.

3.2 Why SBA Methods Fail in MBA: The Exclusion Problem

Traditional SBA methods fail in MBA settings because multiple attackers compete for overlapping neural activation pathways. OOD backdoor methods typically rely on redundant neurons to create specialized activation patterns for poisoned samples.

When multiple independent attackers target different classes, these activation pathways conflict. The strongest attacker dominates the shared neurons, while weaker attacks are suppressed — a phenomenon referred to as "infighting."

The paper's t-SNE visualizations show that attacks such as Vanilla, PGD, Neurotoxin, and Chameleon produce poisoned samples clustered together outside the clean target distributions, leading to pathway competition.

3.3 Effective ID Mapping Construction via Adversarial Adaptation

Mirage resolves infighting by constructing In-Distribution (ID) mappings. Instead of forcing poisoned samples into separate OOD pathways, Mirage aligns poisoned samples with the clean feature distribution of the target class.

A lightweight OOD detector is trained using the frozen feature extractor of the current global model. The detector distinguishes poisoned samples from clean target-class samples.

The trigger is then optimized adversarially to fool the detector. This min-max optimization pushes poisoned samples closer to the clean target distribution, allowing the backdoor to use the same activation pathway as legitimate target-class samples.

3.4 Persistent ID Mapping Enhancement via Constrained Optimization

Adversarial adaptation alone is not enough for long-term persistence. Because the global model Since the global model continuously changes during FL training, the target feature distribution also shifts over time. Adversarial adaptation alone may place poisoned samples near the boundary of the target distribution, making them unstable.

Mirage improves persistence using two additional losses:

- a cosine similarity loss that separates poisoned features from original clean features
- a cross-entropy loss that pulls poisoned features toward the target class

Together, these losses tighten the poisoned distribution within the target feature space and improve long-term attack persistence.

3.5 Complete Attack Procedure

For every selected federated round, each attacker independently performs the following steps:

1. Construct poisoned and clean datasets
2. Train the OOD detector using the global model feature extractor
3. Optimize the trigger using adversarial adaptation and enhancement losses
4. Train the poisoned local model and upload updates to the server

No communication or coordination occurs between attackers at any stage.

4. Implementation Details

4.1 Codebase and Environment

Our implementation is based on the official Mirage codebase (github.com/NUAA-SmartSensing/Mirage) and developed in Python using PyTorch. The modified implementation, including dataset and configuration changes, is maintained at github.com/tanikaaaa/Mirage. Experiments were conducted on a MacBook Air using PyTorch's MPS backend.

Key dependencies include PyTorch, PyYAML, torch_dct, and POT for the AFDM extension. A dedicated virtual environment was used for dependency management.

4.2 Federated Learning Configuration

The FL setup follows the paper's default protocol. Table 1 summarizes the key configuration parameters.

Table 1: Federated Learning Configuration

Dataset	Benign Pre-train	Attack Window	Total Clients	Adversaries (M=10)
CIFAR-10	2000 rounds	100 rounds	100	3 (targets: 0, 1, 2)
CIFAR-100	2000 rounds	100 rounds	100	3 (targets: 0, 1, 2)
GTSRB	2000 rounds	100 rounds	100	3 (targets: 0, 1, 2)

Additional parameters: SGD optimizer with learning rate 0.01, batch size 64 for CIFAR-10/CIFAR-100 and 32 for GTSRB, Blend trigger pattern across all attacks, Dirichlet sampling (α = 1) for non-IID data distribution, and 12.5% local poisoning rate per attacker. The attack window opens at round 2000 and runs for 100 rounds.

4.3 Key Code Modifications

The primary modification we made to the official codebase was in the attacker client class (mirage_client.py). The trigger optimization loop required adjusting the OOD detector update frequency and adding CPU fallback for MPS-incompatible operations. A representative excerpt of the trigger optimization step is shown below:

```
# Trigger adversarial adaptation loop
for epoch in range(E):
    D_det = build_detector_dataset(D_local, delta)
    theta_det = train_detector(D_det, global_model.feature_extractor)
    L_det = BCE_loss(theta_det(x + delta), label=IN_DIST)
    L_enh = CE_loss(global_model(x+delta), y_target) +
             cos_sim(feat(x), feat(x+delta))
    delta = delta - lr * grad(L_det + L_enh, delta)  # PGD step
```

The YAML configuration required adding undocumented keys not present in the official README. For example, discriminator_train_samples_pre_class controls how many clean samples per class are used to train the OOD detector. Setting this too low caused the detector to be inaccurate; too high caused memory issues on MPS. Through experimentation we settled on 50 samples per class as a stable value.

4.4 Technical Challenges

Several implementation challenges were encountered during development. The official Mirage codebase required undocumented YAML parameters, causing difficult runtime debugging. Apple Silicon’s MPS backend lacked support for certain trigger optimization operations, requiring selective CPU fallback. Trigger optimization was computationally expensive, with each FL round taking 10–15 minutes, making full-scale reproduction impractical on limited hardware. Additionally, checkpoint resume logic between benign pretraining and attack phases required manual corrections for stable continuation.

5. Experimental Results

5.1 Evaluation Metrics

Clean Accuracy (ACC) measures the global model's performance on unmodified test inputs - a proxy for attack stealthiness, since a large accuracy drop would alert defenders. Attack Success Rate (ASR) is the fraction of triggered test inputs misclassified as the attacker's target class. GAP measures the difference between the maximum and minimum ASR across the three independent attackers; lower GAP indicates more uniform attack effectiveness. Lifespan measures how many additional FL rounds the backdoor remains active (ASR above threshold) after the attack window closes.

5.2 Main Results

Table 2 presents our reproduced results. The asterisk (*) indicates abbreviated pretraining due to hardware constraints.

Table 2: Reproduced Results (No Defense Setting)

Dataset	Model	Clean Accuracy (ACC ↑)	Attack Success Rate (ASR ↑)	Notes
CIFAR-10	ResNet-18	**92.45%**	**99.84% avg (99.93 / 99.97 / 99.63)**	Checkpoint
CIFAR-100	ResNet-18	64-65%*	29-42%*	Abbrev. pretrain
GTSRB	ResNet-18	95.17%*	64.22%*	Abbrev. pretrain

For CIFAR-10, a pretrained 2000-round checkpoint provided by the authors allowed us to run only the 100-round attack phase. Our results - **92.45% ACC and 99.84% mean ASR (99.93%, 99.97%, 99.63% per attacker) - closely match the paper's reported 92.16% ACC and 99.54% ASR, confirming correct implementation.**

For CIFAR-100 and GTSRB, no pretrained checkpoints were available, requiring benign pretraining before launching the attack phase. Due to hardware limitations (MacBook Air, Apple Silicon MPS), each FL round required 10-15 minutes, making the full 2100-round pipeline impractical. Approximately 300–500 rounds of pretraining were completed before running the attack phase.

The lower results on these datasets reflect the undertrained global model, which weakens the OOD detector and ID mapping construction. Despite this, GTSRB still achieved 64.22% ASR, confirming successful attack behavior. CIFAR-100 achieved 29-42% ASR across attackers, with one attacker reaching 29% even on the partially trained model. With full pretraining, all datasets are expected to exceed 95% ASR as reported in the paper.

5.3 Comparison Against Baseline Attacks

Table 3 compares Mirage against five baseline methods in the MBA setting on CIFAR-10 without any defense.

Table 3: Baseline Comparison on CIFAR-10 (No Defense, MBA Setting)

Method	ACC (%)	ASR (%)	GAP (%)	Observation
Vanilla	91.76	31.88	87.68	Severe exclusion — only dominant attacker succeeds
PGD	91.52	31.25	92.78	Same issue as Vanilla; exclusion unresolved
Neurotoxin	92.03	52.99	90.29	Redundant neurons help marginally; still excluded
Chameleon	92.35	39.65	79.98	Partial improvement; still high exclusion
A3FL	91.73	99.52	0.23	Strong baseline; but OOD signature makes it detectable
Mirage (Ours)	92.25	99.84	0.27	Best overall; ID mapping bypasses OOD defenses

The results clearly confirm the exclusion problem. Vanilla, PGD, and Chameleon achieve only 31–40% mean ASR with GAP values above 79%, indicating that one attacker dominates while the others are largely suppressed. Neurotoxin improves to 53% mean ASR through redundant neuron utilization, but still suffers from a very high GAP of 90%.

A3FL is the strongest baseline (99.52% ASR, 0.23 GAP) because its adversarial unlearning mechanism creates separate OOD activation paths for different attackers. However, this stronger OOD behavior also makes it more vulnerable to BackdoorIndicator. Mirage achieves comparable performance (99.84% ASR, 0.27 GAP) while maintaining an ID feature signature that significantly reduces OOD detectability.

5.4 Evaluation Under Defense Methods

Table 4 summarizes attack performance under six SOTA defense methods on CIFAR-10.

Table 4: Defense Evaluation on CIFAR-10 (Mirage vs. A3FL)

Defense	A3FL ASR (%)	Mirage ASR (%)	Mirage ACC (%)	Defense Type
No Defense	99.52	99.54	92.16	Baseline
BackdoorIndicator	70.33	93.46	91.10	OOD-based detection
Multi-Krum	78.15	92.30	92.10	Byzantine-robust aggregation

Defense	A3FL ASR (%)	Mirage ASR (%)	Mirage ACC (%)	Defense Type
FLAME	79.40	95.90	92.36	DP-based noise injection
FoolsGold	95.30	97.35	91.32	Similarity-based reweighting
DeepSight	96.56	97.39	91.28	Model inspection

The most significant result is the BackdoorIndicator row. This defense - the strongest available for SBA - reduces A3FL's ASR from 99.52% to 70.33% on CIFAR-10. Against Mirage, the same defense achieves only a reduction to 93.46%. This gap directly validates the paper's core thesis: ID mappings bypass OOD-based detection because poisoned samples follow clean activation pathways throughout the network. BackdoorIndicator's OOD detection heuristic simply does not fire for ID-mapped backdoors.

FoolsGold and DeepSight are largely ineffective against both methods because they target gradient similarity among colluding attackers. Non-cooperative MBA attackers naturally produce diverse gradients, defeating similarity-based reweighting. Multi-Krum and FLAME provide some mitigation by modifying aggregation or injecting noise, but Mirage sustains ASR above 92% even under these defenses.

5.5 Attack Persistence

According to Li et al. [1], Mirage maintains ASR above 90% for over 900 rounds after the attack window closes, matching A3FL in persistence. In contrast, Vanilla, PGD, Neurotoxin, and Chameleon rapidly degrade as their OOD mappings are erased by benign updates. An ablation removing constrained optimization significantly reduces persistence, confirming the importance of tightly embedding poisoned samples within the target distribution.

This behavior is consistent with our CIFAR-10 results, where all three attackers maintained ASRs above 99% at the end of the attack phase while preserving 92.25% clean accuracy, indicating a stable and deeply embedded ID mapping.

6. Analysis and Discussion

6.1 Why ID Mapping Resolves Infighting

Mirage’s key insight becomes clear when compared to the failure of traditional OOD backdoor mappings. OOD attacks rely on specialized high-activation neurons, causing multiple attackers to compete for overlapping neural pathways. In MBA settings, the strongest attacker dominates these shared pathways while weaker attacks are suppressed, creating the “infighting” problem.

ID mappings avoid this conflict entirely. After trigger optimization, poisoned samples follow the same activation pathway as clean samples of the target class. Since different target classes naturally use different feature pathways, multiple attackers can coexist without interfering with one another.

This behavior was also observed in our implementation through t-SNE visualizations: baseline methods produced a shared OOD poisoned cluster, while Mirage embedded each attacker's poisoned samples directly within the corresponding target-class feature distribution.

6.2 Defense Gap and the MBA Threat Model

A key takeaway from this project is that most existing FL defenses are designed specifically for SBA settings and perform poorly against MBA attacks. FoolsGold relies on gradient similarity among colluding attackers, which is ineffective for independent adversaries. BackdoorIndicator depends on detecting OOD feature signatures, making it ineffective against Mirage's ID mappings. Aggregation-based defenses such as Multi-Krum and FLAME are also less effective against the low-variance poisoned updates generated by Mirage.

6.3 Limitations of Our Reproduction

Our reproduction has two main limitations. First, full CIFAR-100 and GTSRB experiments required 2000-round pretraining, which exceeded our hardware constraints; therefore, these results reflect abbreviated pretraining. Second, we could not independently run the full lifespan experiment extending to round 3000 due to computational limitations.

These are practical infrastructure constraints rather than algorithmic limitations. The CIFAR-10 experiments, which used the provided pretrained checkpoint, closely reproduced the paper's reported performance and confirmed correct implementation of the attack framework.

7. Proposed Novelty: Adaptive Frequency-Domain Mirage (AFDM)

7.1 Motivation

Mirage optimizes triggers directly in pixel space using PGD. While effective, pixel-space triggers concentrate perturbations in specific spatial regions, making them potentially detectable through spatial anomaly analysis or perceptual quality metrics.

Modern CNNs naturally learn hierarchical spectral features, with early layers responding strongly to edges and frequency patterns. This motivates optimizing triggers in the frequency domain, where perturbations are represented as spectral coefficients instead of raw pixel values. Such triggers can align more naturally with learned feature representations while distributing perturbation energy across frequency bands in a less visually detectable manner.

7.2 Core Design

AFDM applies triggers in the frequency domain using the Discrete Cosine Transform (DCT). The pipeline follows:
Image → DCT → Spectral Coefficient Modification → IDCT → Poisoned Image.

Instead of optimizing a pixel-space patch, the trigger is represented as spectral coefficients. The original Mirage adversarial adaptation and constrained optimization losses are retained, while gradients are backpropagated through the IDCT operation to update the frequency-domain trigger.

AFDM additionally introduces:

1. **Spectral Attention** to focus optimization on important frequency bands,
2. **Adaptive Frequency Modulation** to dynamically scale perturbation strength based on ID mapping quality,
3. **Spectral Smoothness Loss** to reduce abrupt spectral discontinuities and improve stealthiness.

7.3 Implementation Status and Preliminary Exploration

AFDM infrastructure was implemented using the `torch_dct` library for differentiable DCT operations, with gradient propagation supported through PyTorch autograd. However, full-scale AFDM training was not completed within the project timeline due to the added computational cost of DCT/IDCT operations during trigger optimization.

Key open research questions include whether frequency-domain triggers improve ID mapping quality, enhance resistance to spatial anomaly detectors, and balance spectral smoothness with attack effectiveness. Evaluating AFDM against frequency-aware defenses would be a meaningful future research direction.

8. Conclusion

This project successfully reproduced the Mirage CVPR 2025 framework for multi-label backdoor attacks in federated learning and provided practical insight into the challenges of implementing large-scale FL systems. Setting up federated simulations, handling YAML configuration issues, managing checkpoints, and working around MPS backend limitations highlighted the gap between published research and real-world implementation.

The key insight of Mirage is that replacing Out-of-Distribution (OOD) mappings with In-Distribution (ID) mappings fundamentally changes the behavior of backdoor attacks. By aligning poisoned samples with clean target-class feature pathways, Mirage eliminates inter-attacker exclusion without requiring coordination and naturally bypasses OOD-based defenses.

A major finding from this work is the current defense gap in the MBA setting. Existing defenses such as FoolsGold, BackdoorIndicator, Multi-Krum, and FLAME were primarily designed for SBA attacks and remain ineffective against multiple simultaneous ID-mapped backdoors.

In addition, we proposed Adaptive Frequency-Domain Mirage (AFDM), a novel extension that shifts trigger optimization into the frequency domain using DCT-based spectral manipulation.

Combining ID mapping with spectral trigger optimization could provide a stronger multi-layer evasion strategy against both feature-space and spatial anomaly detection methods, making it a promising direction for future research.

References

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